

Orbit and eye movements

Localise the structure before naming the lesion

Orbital routes walls, openings and vessels

Walls and adjacent spaces

Wall	Bones / useful relationship
Roof	Frontal and lesser sphenoid wing; cranial cavity above
Floor	Mainly maxilla, with zygomatic and palatine; maxillary sinus below
Medial	Maxilla, lacrimal, ethmoid and sphenoid; thin ethmoid wall
Lateral	Zygomatic and greater sphenoid wing; stronger wall

Two routes that separate syndromes

Opening	Key contents
Optic canal	CN II and ophthalmic artery
Superior orbital fissure	CN III, IV, VI; V1 branches; ophthalmic veins

The common tendinous ring

The four recti arise from this ring near the orbital apex. CN III divisions, CN VI and nasociliary V1 pass through its enclosed portion of the superior orbital fissure; frontal V1, lacrimal V1 and CN IV travel outside the ring. The optic canal also lies within the ring region.

Translate anatomy into localisation

- The ophthalmic artery arises from the internal carotid. Its central retinal branch supplies inner retina; ciliary vessels supply choroid.
- Orbital veins connect facial veins and cavernous sinus. Valveless channels allow infection to spread with pressure gradients.
- Ethmoid infection can cross the thin medial wall. Impaired eye movements suggest infection beyond the eyelid.
- Ocular motor palsies with V1 loss fit the superior orbital fissure. Added optic-nerve dysfunction suggests orbital apex or optic canal involvement.

Six muscles action is not testing position

Muscle actions

Muscle / nerve	Actions
Medial rectus / III	Adduction
Lateral rectus / VI	Abduction
Superior rectus / III	Elevation, intorsion, adduction
Inferior rectus / III	Depression, extorsion, adduction
Superior oblique / IV	Intorsion, depression, abduction
Inferior oblique / III	Extorsion, elevation, abduction

How to isolate vertical movement

ABDUCTED eye

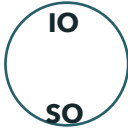
Up in abduction



Down in abduction

ADDUCTED eye

Up in adduction



Down in adduction

ABDUCT, then look up or down: superior/inferior rectus. ADDUCT, then look up or down: inferior/superior oblique. Labels refer to the tested eye, not the observer.

Why the H test works

A muscle can have several actions in primary gaze. Horizontal positioning changes its alignment with the globe: vertical recti contribute most effectively to vertical testing in abduction; obliques contribute most effectively in adduction. The test position therefore does not simply copy the muscle’s full action list.

Paired gaze and laterality

- Down-right gaze: right inferior rectus plus left superior oblique. Up-right gaze: right superior rectus plus left inferior oblique.
- Version moves both eyes in the same direction. Convergence adducts both eyes towards a near target; vergence describes opposite-direction binocular movements.

Weakness, restriction and eyelid clues

Pattern and structure

Pattern	Interpretation
Ptosis, down-and-out eye	CN III: most ocular muscles plus levator impaired
Impaired abduction	Ipsilateral lateral rectus / CN VI dysfunction
Impaired depression in adduction	Ipsilateral superior oblique / CN IV dysfunction
Globe cannot elevate; tight inferior rectus	Mechanical restriction, rather than a weak elevator

Explain the resting position

In a complete CN III palsy, functioning lateral rectus abducts the eye and superior oblique contributes depression. Levator weakness produces marked ptosis; parasympathetic involvement can leave a dilated pupil.

Eyelid muscles tell different stories

Muscle	Supply / action
Levator palpebrae superioris	CN III superior division; lifts upper lid
Superior tarsal muscle	Sympathetic; assists lid elevation
Orbicularis oculi	CN VII; closes lids and assists tear drainage

Reason through the limitation

- CN III superior division supplies superior rectus and levator. The inferior division supplies the other III muscles and preganglionic parasympathetic fibres.
- A fibrotic inferior rectus restricts elevation, even when superior rectus strength is normal.
- Thyroid eye disease often restricts recti, especially inferior and medial rectus. Orbital-fracture entrapment is another mechanical cause.
- CN VI can be affected by raised intracranial pressure, so an isolated abduction deficit does not always pinpoint a local pontine lesion.
- Mild ptosis with miosis favours sympathetic dysfunction; failure to close the eye points towards facial motor weakness.

Corneal reflex and the tear pathway

Corneal blink arc

- Corneal touch → nasociliary branch of V1
- ↓
- Brainstem trigeminal sensory pathway
- ↓
- Bilateral facial motor nuclei → CN VII
- ↓
- Orbicularis oculi contraction in both eyes

Test each cornea, watch both lids

Lesion	Expected response
Right V1 afferent failure	Right touch: neither blinks. Left touch: both blink
Right VII motor failure	Either touch: left lid blinks; right lid does not

Tear production and drainage

- Secretomotor: CN VII → greater petrosal nerve → nerve of pterygoid canal → pterygopalatine ganglion
- ↓
- Postganglionic fibres travel via V2 and then lacrimal V1 to the gland
- ↓
- Drainage: puncta → canaliculi → lacrimal sac
- ↓
- Nasolacrimal duct → inferior nasal meatus

Tear-film components

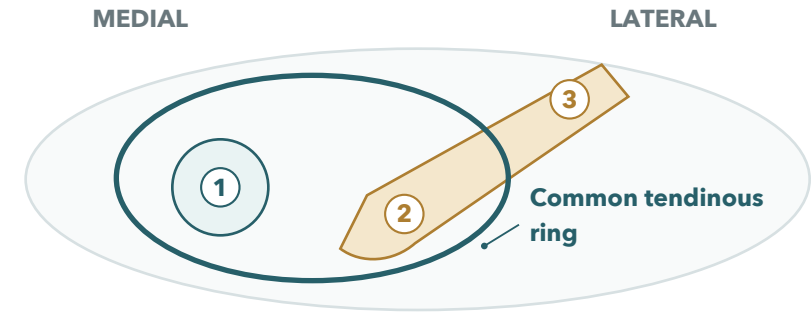
Component	Source / contribution
Lipid at the surface	Mainly Meibomian glands; reduces evaporation
Aqueous	Lacrimal and accessory glands; fluid, nutrients and defence
Mucin at the ocular surface	Conjunctival goblet cells; promotes surface wetting

Two distinctions that save marks

The lacrimal nerve is a V1 sensory branch that carries postganglionic secretomotor fibres; those fibres originate in the facial parasympathetic system. Lacrimal-gland secretion and tear drainage are separate processes: normal secretion can coexist with overflow tearing when outflow or the eyelid pump fails. The gland sits superotemporally, whereas the puncta and sac lie medially.

Picture the apex and paired gaze

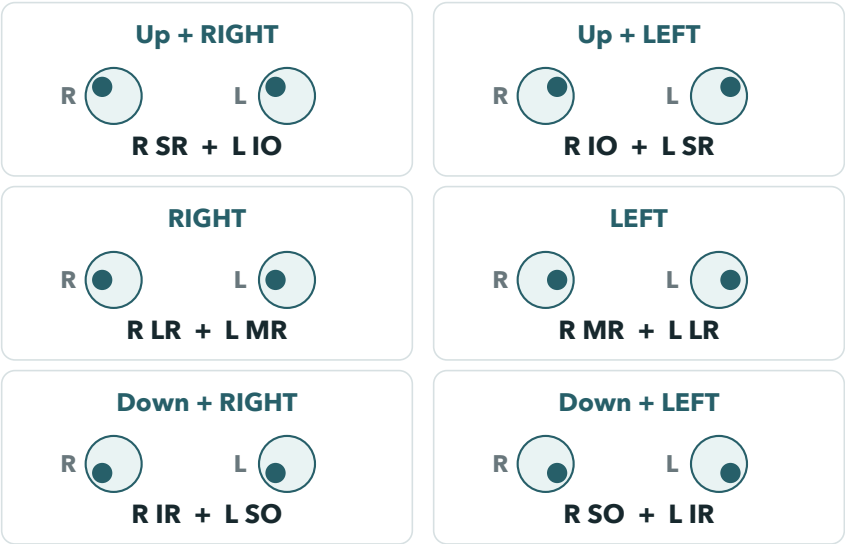
1 The ring divides the nerve routes



- ① **Optic canal**
CN II + ophthalmic artery
- ② **SOF: inside the ring**
CN III divisions, VI, nasociliary V1
- ③ **SOF: outside the ring**
CN IV, frontal V1, lacrimal V1

Original map of the key apex relationships, not a surgical scale drawing. The optic canal lies medial to the SOF. Venous pathways are omitted for visual clarity.

2 Six gaze positions, two eyes



Original frontal-view schematic: patient right appears on the left. R/L label the patient's eyes. Principal muscles are shown: LR uses VI, SO uses IV, and the others III.

Use the pictures to localise

Multiple motor deficits with V1 loss suggest the SOF region. Optic-nerve dysfunction adds the optic canal/apex. Identify the eye's horizontal position before choosing its vertical muscle.